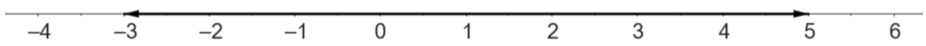
IMAGINARY NUMBERS ARE REAL

Most of us are introduced to numbers as tools for counting when we learn how to use the *counting numbers* $1, 2, 3, \ldots$ to answer 'how many?' We learn to use decimal notation and what 321 means. We practice adding and subtracting, and then multiplying and dividing. If we subtract 10 from 3, we are told that the answer is a new kind of number: the negative number -7. The integers consist of all the counting numbers together with their negatives and zero. If we divide 10 by 3 getting 3 remainder 1, we are told that the answer is another new kind of number: the fraction $10/3$. Fractions are also called rationals (from ratio).



We learn that the integers and rationals live on the number line and that any other point on the line is a number too. We learn how to write fractions in the decimal notation and where on the number line they go. By now, we're comfortable using numbers not just to count things, but to, for example, measure things like distances and volumes. There are some points on the line that do not correspond to fractions, but we have a strong sense that they nevertheless correspond to *some* number. For example, we think there is a number somewhere around 1.4 that when multiplied by itself gives 2. We call it $\sqrt{2}$ – the square root of 2 – and even though it isn't a ratio of integers (it's not rational) and we can't write out its decimal expansion in full, most of us are happy thinking of it as just as real as the numbers for which we can.

Many people feel like the square root of -1 on the other hand has a status more like: *"the largest prime number"* or *"the smallest number not definable in less than 15 words"*. After all, any positive number squared is positive and any negative number squared is also positive, so no number at all can square to -1, so the argument goes. In fact numbers that square to negatives have unfortunately become known as *imaginary numbers* while the ones on the line are known as *the real numbers*.

It is certainly true that none of the positive or negative numbers on the familiar number line square to -1. But I want to suggest that the argument just given for why no number is the square root of -1 is quite analogous to each of the following.

- Any number when added to 10 gives a result that is at least 10. But 3 is less than 10. Therefore there is no number that when added to 10 gives 3.
- Any number when multiplied by 3 gives an answer that is a multiple of 3. But 10 is not a multiple of 3. Therefore there is no number that when multiplied by 3 gives 10.
- Any number when squared is equal to a^2/b^2 for some integers a and b. But there are no integers a and b such that $2b^2 = a^2$. Therefore there is no number that when squared is equal to 2.

In each case, what is really being shown is that there is no number *of a certain kind* with the desired property. In each case there were good reasons to introduce new kinds of numbers. Good enough that we learned to think and talk about the new kinds as just as real as the old. The numbers -7, $10/3$ and $\sqrt{2}$ have more *ontological oomph* than 'the smallest number not definable in less than 15 words'. What follows is an attempt at showing $\sqrt{-1}$ does too.

The plan is to look at four case studies in which square roots of negative numbers play a key role. After that, we'll introduce a little theory about $\sqrt{-1}$ in order to go back and better understand them in more detail.

1. Example 1: Fibonacci

Consider the sequence

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots.$$

Let f_n denote the n^{th} term. It is generated by the rule $f_{n+2} = f_{n+1} + f_n$ for $n = 0, 1, 2, \ldots$ together with the initial values $f_0 = 1$ and $f_1 = 1$. We can calculate the n^{th} term in the sequence using the following formula

$$f_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

A proper explanation for why this formula is true is given in Appendix A, but for our present purposes it is enough to get a rough idea. The idea is this. If we ignore the actual starting values for now, and only focus on the rule $f_{n+2} = f_{n+1} + f_n$, then it's easy enough to see that $f_n = x^n$ satisfies this rule provided x satisfies the quadratic equation $x^2 = x + 1$. Recall the quadratic formula that says the solutions to $ax^2 + bx + c = 0$ are given by $\frac{-b\pm\sqrt{b^2-4ac}}{2a}$. So the values of x we're after are $\frac{1\pm\sqrt{5}}{2}$, and this explains the appearance of these numbers in the formula. The actual formula is then a variant of this that can be made to also satisfy the initial values $f_0 = 0$ and $f_1 = 1$.

What if the resulting quadratic equation didn't have any real solutions though? Consider for example the sequence u_n which is generated by the rule $u_{n+2} = 2u_{n+1} - 3u_n$ and the starting values $u_0 = 1$ and $u_1 = 1$

$$1, 1, -1, -5, -7, 1, 23, 43, 17, -95, \ldots.$$

The same approach this time has us wanting to solve the quadratic $x^2 = 2x - 3$ and the quadratic formula is telling us the solutions are $x = 1 + \sqrt{-2}$ and $x = 1 - \sqrt{-2}$. But now there seems to be a problem because we need to take the square-root of a negative number. At this point it might be tempting to conclude that we must have gone wrong somewhere. We could respond with "the formula we get involves square roots of negative numbers, but there are no such numbers and therefore the formula is (in some sense at least) 'wrong'. It has failed to produce for us a formula for the n^{th} term in the sequence as it did for the previous example." In fact there are square roots of negative numbers, it's not wrong and it does still work.

Let's just drop the supposition that imaginary numbers are not real and suppose instead there *is* some number $\sqrt{-2}$ with the property that $(\sqrt{-2})^2 = -2$. Suppose that it's possible to think of and treat $\sqrt{-2}$ as some new kind of number that we can add to our familiar numbers like 2. Going through the same argument would yield the formula

$$u_n = \frac{1}{2}\left((1+\sqrt{-2})^n + (1-\sqrt{-2})^n\right).$$

Does this actually work? Yes, it does! Take $n = 8$ for example. We can see above that $u_8 = 17$. It's easy enough to apply the formula by hand if we use that fact that $(1+\sqrt{-2})^8 =$

$(((1+\sqrt{-2})^2)^2)^2$ so that we just have to square three times and use the fact that $(\sqrt{-2})^2 = -2$.

$$\begin{aligned}(((1+\sqrt{-2})^2)^2)^2 &= ((-1+\sqrt{-2})^2)^2\\ &= (-7-4\sqrt{-2})^2\\ &= 17+56\sqrt{-2}\end{aligned}$$

Similarly $(((1-\sqrt{-2})^2)^2)^2 = 17-56\sqrt{-2}$ and the formula reproduces

$$u_8 = \frac{1}{2}(17+56\sqrt{-2}+17-56\sqrt{-2}) = 17$$

exactly as hoped for. Later we will see the connection between complex number multiplication and rotations which will allow us to very quickly derive the equivalent formula

$$\frac{1}{2}\left((1+\sqrt{-2})^n + (1-\sqrt{-2})^n\right) = (\sqrt{3})^n \cos(n \arccos(\frac{1}{\sqrt{3}})).$$

2. Example 2: Cardano

In the previous example we made use of the quadratic formula to solve equations of the form $ax^2+bx+c=0$. This example is about the cubic equation

$$x^3+cx+d=0.$$

The general cubic equation would be $ax^3+bx^2+cx+d=0$ but to simplify the algebra we're going to assume that $a=1$ and $b=0$. Some very clever algebraic manipulations (see Appendix B for details) lead to the following formula for x in terms of c and d,

$$x = \sqrt[3]{\frac{-d}{2}+\sqrt{\frac{d^2}{4}+\frac{c^3}{27}}} + \sqrt[3]{\frac{-d}{2}-\sqrt{\frac{d^2}{4}+\frac{c^3}{27}}}.$$

To see how this works we can use it to solve $x^3+6x-20=0$. It tells us one solution is

$$x = \sqrt[3]{10+\sqrt{108}} + \sqrt[3]{10-\sqrt{108}}$$

and we can check on a calculator that

$$x = \sqrt[3]{20.392305} + \sqrt[3]{-0.392305} = 2.7320508 - 0.7320508 = 2$$

So we have found the solution $x=2$ and in the process the curious identity

$$\sqrt[3]{10+\sqrt{108}} + \sqrt[3]{10-\sqrt{108}} = 2.$$

Let's now try with $x^3-15x-4=0$. This time the formula gives

$$x = \sqrt[3]{2+\sqrt{-121}} + \sqrt[3]{2-\sqrt{-121}}$$

But now we have the same 'problem' with our formula as in Example 1. It's telling us we need to take the square-root of a negative number and we might again be tempting to conclude that we must have gone wrong somewhere. Even if we could treat $\sqrt{-121}$ as some new kind of number like we did in the previous example, what could it mean to take a cube root of $2+\sqrt{-121}$? Is this yet another new kind of number? Maybe what's going on is that there are no real solutions. But we know there is because if we plot the graph of $y=f(x)$ we can clearly see it crosses the x-axis. In fact there will always be at least 1 real root no matter what c and

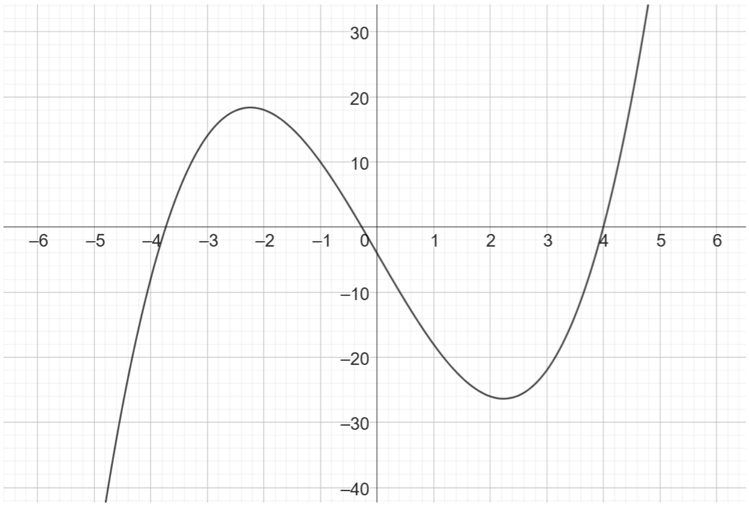


FIGURE 1. $y = x^3 - 15x - 4$

d are. Well maybe the formula simply doesn't work when $d^2/4 + c^3/27 < 0$. In fact it does still work if we know how to use it.

As before, let's just drop the supposition that $\sqrt{-121}$ isn't real. Suppose instead there is some new kind of number that we can add to our familiar numbers like 2 with the property it's square is -121. What about $\sqrt[3]{2+\sqrt{-121}}$? Later on we'll see that we need to be a bit careful what talking about cube roots[1] but for now let's simply note the following calculation

$$(2+\sqrt{-1})^3 = (2+\sqrt{-1})(3+4\sqrt{-1}) = 2+11\sqrt{-1}$$

so that if $\sqrt{-121} = \sqrt{11^2\cdot(-1)} = 11\sqrt{-1}$ then $2+\sqrt{-1}$ is a cube root of $2+\sqrt{-121}$. A very similar calculation shows that $(2-\sqrt{-1})^3 = 2-11\sqrt{-1}$. Now if we interpret these expressions with $\sqrt{-1}$ as the cube roots in our formula then it yields the solution

$$\sqrt[3]{2+\sqrt{-121}} + \sqrt[3]{2-\sqrt{-121}} = 2+\sqrt{-1}+2-\sqrt{-1} = 4.$$

and it is easy to verify that $x = 4$ is indeed a solution to $x^3 - 15x - 4 = 0$.

So the formula still works if we are able to make sense of $\sqrt{-1}$ as a legitimate number and know how to calculate with it. But how did we know that $2+\sqrt{-1}$ was going to cube to $2+\sqrt{-121}$? Could we have worked this out from the formula? Yes, we'll see later after we learn a little more about the behaviour of $\sqrt{-1}$ how to find the cube root of $2+\sqrt{-121}$. We'll also see how the formula includes the other two roots we can see from the graph around -3.7 and -0.3.

3. EXAMPLE 3: INTEGRATING $\frac{1}{1+x^2}$

Consider the area under the curve $y = \frac{1}{1-a^2x^2}$

$$\int \frac{1}{1-a^2x^2}dx.$$

[1]This is very similar to how we need to be careful when talking about square roots because we could mean the positive or negative square root. Just like there are two square roots, when we allow for complex roots there are 3 possible cube roots, as we shall see.

A standard approach to carrying out the integration is with partial fractions by making use of the factorisation $1 - a^2x^2 = (1 - ax)(1 + ax)$ to first write

$$\frac{1}{1-a^2x^2} = \frac{1}{2a}\left(\frac{1}{1-ax} + \frac{1}{1+ax}\right).$$

The two terms can then be integrated to get

$$\int \frac{1}{1-a^2x^2}dx = \frac{1}{2a}(\log(1+ax) - \log(1-ax)).$$

But what if we instead wanted the integral of $\frac{1}{1+x^2}$? This time $1 + x^2$ doesn't have real roots and cannot be factored using reals numbers. But what is stopping us from setting $a = \sqrt{-1}$? Wouldn't it be nice if we could just go ahead and use the same formula? Well that would tell us

$$\int \frac{1}{1+x^2}dx = \frac{1}{2\sqrt{-1}}(\log(1+\sqrt{-1}x) - \log(1-\sqrt{-1}x))$$

and surely that cannot be right. We know the answer is the area under the curve $y = 1/(1+x^2)$, which is just some real number. In fact, this calculation *is* valid and that *is* the correct answer!

Just before we move to our final example though, here's another way of calculating the integral $1/(1 + x^2)$. Let $x = \tan(y)$. Since $\tan(y) = \sin(y)/\cos(y)$ we can use the quotient rule to get $dx/dy = (\sin(y)^2+\cos(y)^2)/(\cos(y)^2) = 1+\tan(y)^2 = 1+x^2$. Therefore $dy/dx = 1/(1+x^2)$ and so

$$\int \frac{1}{1+x^2}dx = \arctan(x).$$

So, was our partial fraction calculation using $\sqrt{-1}$ wrong after all, or is it *really* true that

$$\arctan(x) = \frac{1}{2\sqrt{-1}}(\log(1+\sqrt{-1}x) - \log(1-\sqrt{-1}x))?$$

Indeed it is. We'll see later how it can be arrived at by other means.

4. Example 4: The surface area of an ellipsoid

Our final example involves calculating the surface area of the shape traced out by an ellipse when rotated on one of its axes, a kind of ellipsoid. The general formula for the area traced out by rotating the graph of the function $y(x)$ around the x-axis is

$$A = 2\pi \int_a^b y(x)\sqrt{1+y'(x)^2}dx$$

As a warm up let's us it to calculate the area of a sphere of radius r. This sphere is generated by rotating a circle of radius r centred at 0 around the x-axis. The equation of the circle is $y^2 + x^2 = r^2$. Then $dy/dx = -x/y$, and the formula tells us the area of the sphere is

$$A = 2\pi \int_{-r}^{r} y\sqrt{1+(-x/y)^2}dx = 2\pi \int_{-r}^{r} \sqrt{y^2+x^2}dx = 2\pi r \int_{-r}^{r} dx = 4\pi r^2.$$

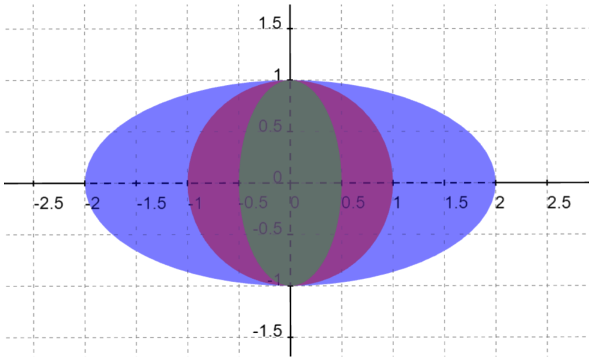


FIGURE 2. Example ellipses with $a = 1$ and $b = 1/2, 1, 2$

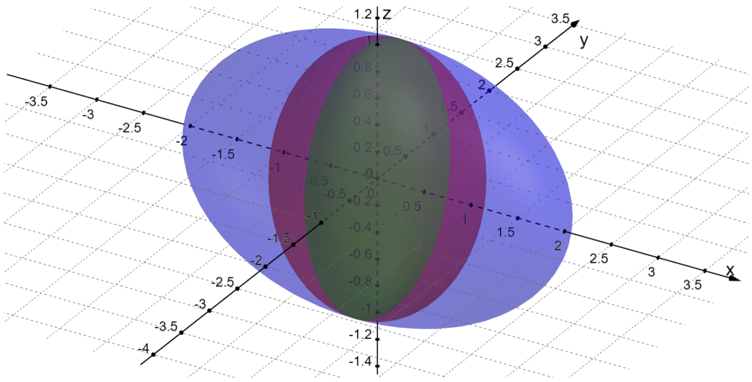


FIGURE 3. Example ellipsoids with $a = 1$ and $b = 1/2, 1, 2$

Now for the ellipsoid we'll rotate the ellipse defined by $x^2/a^2 + y^2/b^2 = 1$. This time $dy/dx = -\frac{b^2x}{a^2y}$ and the area is

$$\begin{aligned} A &= 2\pi \int_0^a y\sqrt{1 + (-\frac{b^2x}{a^2y})^2}dx \\ &= 2\pi \int_0^a \sqrt{y^2 + b^4x^2/a^4}dx \\ &= 2\pi \int_0^a \sqrt{b^2 + x^2(b^4/a^4 - b^2/a^2)}dx \\ &= 2\pi b \int_0^a \sqrt{1 + x^2\frac{b^2 - a^2}{a^4}}dx \end{aligned}$$

We can simplify this a bit by making the substitution $u = x\sqrt{(b^2 - a^2)/a^4}$ if $b > a$ and $u = x\sqrt{(a^2 - b^2)/a^4}$ if $a > b$

$$A = 2\pi abc \int_0^c \sqrt{1 + u^2}du \text{ if } b > a \text{ with } c = \sqrt{(b/a)^2 - 1}$$
$$A = 2\pi abc \int_0^c \sqrt{1 - u^2}du \text{ if } a > b \text{ with } c = \sqrt{1 - (b/a)^2}$$

These integrals now look simpler, but it is still not obvious how to evaluate them. They can be done though after some clever substitutions. The following formula are deduced in Appendix C.

$$\int_0^x \sqrt{1+u^2}du = \frac{x\sqrt{x^2+1}}{2} + \frac{1}{2}\log(x+\sqrt{x^2+1})$$
$$\int_0^x \sqrt{1-u^2}du = \frac{x\sqrt{1-x^2}}{2} + \frac{1}{2}\arcsin(x)$$

A little algebra then shows that

$$A = \pi ab\left(b/a + \frac{1}{\sqrt{(b/a)^2-1}}\log(b/a+\sqrt{(b/a)^2-1})\right) \text{ if } b > a$$
$$A = \pi ab\left(b/a + \frac{1}{\sqrt{1-(b/a)^2}}\arcsin(\sqrt{1-(b/a)^2})\right) \text{ if } a > b.$$

To simplify a bit further, let's suppose that $a = 1$ and write the area as a function of b

$$A(b) = \pi b\left(b + \frac{1}{\sqrt{b^2-1}}\log(b+\sqrt{b^2-1})\right) \text{ if } b > 1$$
$$A(b) = \pi b\left(b + \frac{1}{\sqrt{1-b^2}}\arcsin(\sqrt{1-b^2})\right) \text{ if } b < 1$$

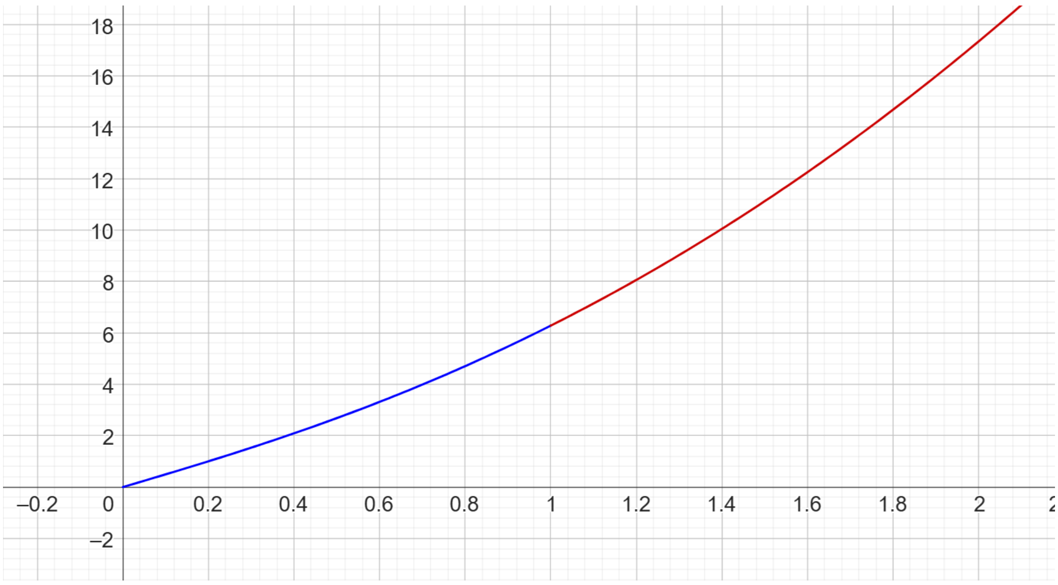


FIGURE 4. Area of ellipsoid as a function of b with $a = 1$

But now this is quite interesting. As the parameter b varies, the shape of the ellipsoid changes. When $b = 1$ it is a sphere. The area is quite obviously a smoothly varying function of b but our calculation has given us two seemingly very different formula for its value. One option is to say that there are no square roots of negative numbers so the first simply doesn't make sense when $b < 1$. Similarly, the second just doesn't apply when $b > 1$. But this would be quite analogous to dismissing our cubic formula from earlier on the grounds that it too involved square roots of negative numbers. We would have missed that it did in fact produce the real solution to the equation.

Alternatively, we can boldly set the two formula equal to each other and write down

$$\frac{1}{\sqrt{x^2-1}}\log(x+\sqrt{x^2-1}) = \frac{1}{\sqrt{1-x^2}}\arcsin(\sqrt{1-x^2})$$

even though at least one of $\sqrt{1-x^2}$ and $\sqrt{x^2-1}$ is negative.

There are definite similarities between this strange equation and the one from the previous example. Both involve square-roots of negative numbers, and both equate an inverse trigonometric function to the natural logarithm function. Indeed it really is possible to make sense of these equalities once we embrace $\sqrt{-1}$ into our system of numbers. They are both manifestations of one of the most famous formulas in all of mathematics, which we will come to. But we are getting ahead of ourselves. Let's start by trying to understand some of the more basic algebraic properties of $\sqrt{-1}$.

5. Basic properties of $\sqrt{-1}$

5.1. **Complex numbers.** To start our understanding of the strange formula introduced in the preceding examples we need to get clearer on some of the basic properties of these new numbers with negative squares. It is helpful to introduce a new symbol i to label one of these with the key property

$$i^2 = -1$$

and then consider all the numbers of the form $a+ib$ where a and b are real numbers. We call a the *real part* and b the *imaginary part*. Two such numbers $a+ib$ and $c+id$ are equal if and only if $a=c$ and $b=d$. The set of all such numbers is called the set of *complex numbers* and we shall take for granted that they behave according to all the same basic arithmetic rules as real numbers do. In particular, the following identities hold for all real a, b, c, d

$$(a+ib)+(c+id) = (a+c)+i(b+d)$$
$$(a+ib)(c+id) = (ac-bd)+i(ad+bc).$$

The second is a consequence of expanding brackets in the familiar way and using $i^2=-1$. It is also useful to know how to divide complex numbers and write the answer in the form $a+ib$ which we can do as follows

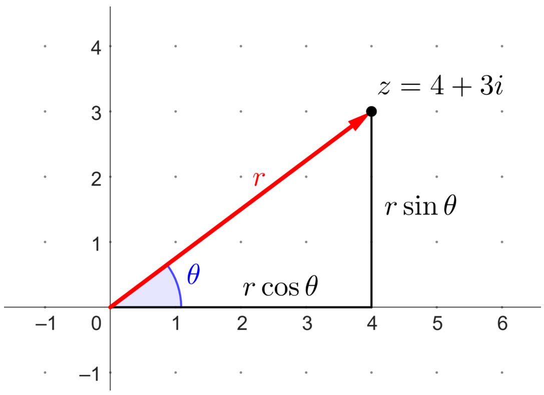
$$\frac{a+ib}{c+id} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = \frac{ac+bd}{c^2+d^2} + i\frac{bc-ad}{c^2+d^2}.$$

Equipped with this new number i we can give meaning to the square roots of negative numbers like $\sqrt{-121}$, $\sqrt{-2}$ we saw before by writing them as a complex numbers in the form $a+ib$. We define $\sqrt{-121} = i\sqrt{121}$ and $\sqrt{-2} = i\sqrt{2}$ etc. The multiplication rule above implies that $(\sqrt{-2})^2 = (i\sqrt{2})^2 = -2$, just as we would want from a number written $\sqrt{-2}$. A word of warning though, because it's possible to get into trouble very quickly with this notation if we're not careful. Consider the following 'proofs' that $1=-1$

$$1 = \sqrt{1} = \sqrt{(-1)^2} = (\sqrt{-1})^2 = -1$$
$$-1 = \sqrt{-1} = \sqrt{-1}\sqrt{\frac{1}{-1}} = \sqrt{-1}\frac{\sqrt{1}}{\sqrt{-1}} = 1$$

The problem is that familiar rules likes $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{1/a} = 1/\sqrt{a}$ don't necessarily apply when a and b are negative.

5.2. **The complex plane.** Part of why complex numbers might feel more abstract and less *real* than real numbers might be that we're so accustomed to *seeing* numbers as positions on the number line. We can represent complex numbers visually too, by associating the number $a+ib$ with the point (a,b) in the Cartesian xy-plane. So i is located at the point $(0,1)$ and real number line is identified with the x-axis and is called the real axis. The y-axis is associated with all the points of the form ib where b is real and is called the imaginary axis. Addition of complex numbers then corresponds to addition of 2-dimensional vectors with real coordinates.



It turns out to be extremely useful (for reasons we are about to see) to describe complex numbers in *polar form*. That is, in terms of their distance from 0 (the origin) and the angle that is formed by the line through the number and 0 and the x-axis. The number $z = a+ib$ is written in polar form as

$$z = r(\cos(\theta) + i\sin(\theta))$$

where $r = \sqrt{a^2+b^2}$ and $\theta = \arctan(b/a)$ depend on z. The angle θ is always measured anti-clockwise from the x-axis and will typically be chosen so that $-\pi < \theta \leq \pi$. The number r is called the *modulus* of z and is written $|z|$, while the angle θ is called the *argument* of z and is written $\arg(z)$.

The following result explains why it is often so useful to write complex numbers in polar form. We've seen that when we add complex numbers we add their real and imaginary parts separately, just like with vectors. This identity explains how the polar coordinates r and θ change when we multiply complex numbers.

Proposition. *Let $z = r(\cos(\theta)+i\sin(\theta))$ and $w = s(\cos(\phi)+i\sin(\phi))$ be two complex numbers written in polar form. The polar form of the product zw is*

$$rs(\cos(\theta+\phi) + i\sin(\theta+\phi)).$$

In particular, when we multiply two complex numbers we multiply their moduli and add their arguments.

Proof. This is a direct consequence of the angle addition formulas[2] for sin and cos

$$\sin(\theta+\phi) = \sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi)$$
$$\cos(\theta+\phi) = \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi).$$

[2]See https://mathworld.wolfram.com/TrigonometricAdditionFormulas.html for a proof.

With these, we simply need to multiply out the brackets and remember that $i^2 = -1$

$$\begin{aligned} zw &= rs(\cos(\theta) + i\sin(\theta))(\cos(\phi) + i\sin(\phi)) \\ &= rs\big(\cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi) + i(\sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi))\big) \end{aligned}$$ □

6. De Moivre's theorem

An immediate consequence of this Proposition is De Moivre's Theorem which says that for all r and θ and all positive integers n

$$(r(\cos\theta + i\cos\theta))^n = r^n(\cos(n\theta) + i\sin(n\theta)).$$

Why? Because the left hand side is just $r(\cos\theta + i\sin\theta)$ multiplied by itself n times so the Proposition tells us that the modulus is just r is multiplied by itself n times (giving r^n) and the argument is θ added to itself n times (giving $n\theta$).

By setting $r = 1$ and substituting θ for $\frac{\theta}{n}$ we get the rather nice

$$\cos(\theta) + i\sin(\theta) = (\cos(\tfrac{\theta}{n}) + i\sin(\tfrac{\theta}{n}))^n.$$

Equating the real and imaginary parts of either side of this equation we can very cleanly generate trigonometric identities giving $\cos(\theta)$ and $\sin(\theta)$ in terms of powers of $\sin(\frac{\theta}{n})$ and $\cos(\frac{\theta}{n})$. The reader may wish to verify that with $n = 4$ for example we get

$$\begin{aligned} \cos(\theta) &= \cos(\tfrac{\theta}{4})^4 - 6\cos(\tfrac{\theta}{4})^2\sin(\tfrac{\theta}{4})^2 + \sin(\tfrac{\theta}{4})^4 \\ \sin(\tfrac{\theta}{4}) &= 4\cos(\tfrac{\theta}{4})^3\sin(\tfrac{\theta}{4}) - 4\cos(\tfrac{\theta}{4})\sin(\tfrac{\theta}{4})^3 \end{aligned}$$

and with $n = 6$ we get

$$\begin{aligned} \cos(\theta) &= \cos(\tfrac{\theta}{6})^6 - 15\cos(\tfrac{\theta}{6})^4\sin(\tfrac{\theta}{6})^2 + 15\cos(\tfrac{\theta}{6})^2\sin(\tfrac{\theta}{6})^4 - \sin(\tfrac{\theta}{6})^6 \\ \sin(\theta) &= 6\cos(\tfrac{\theta}{6})^5\sin(\tfrac{\theta}{6}) - 20\cos(\tfrac{\theta}{6})^3\sin(\tfrac{\theta}{6})^3 + 6\cos(\tfrac{\theta}{6})\sin(\tfrac{\theta}{6})^5 \end{aligned}$$

As n gets bigger and bigger these identities get more and more complicated with more and more terms involving even larger binomial coefficients and powers of $\sin(\frac{\theta}{n})$ and $\cos(\frac{\theta}{n})$. Something rather magical starts to happen though as n continues on its journey to infinity. The formula starts to settle down again until eventually, when $n = \infty$ itself, we're left with something rather special. A formula that we'll see sheds light on the strange identities from Examples 3 and 4.

Before we get to that though, let's use what we have done so far to address some of the outstanding questions raised by Examples 1 and 2.

6.1. The recurrence formula.

Recall our formula from Example 1 for the n^{th} term of our Fibonacci-like sequence $1, 1, -1, -5, -7, 1, 23, 43, 17, -95, \ldots$

$$\frac{1}{2}\Big((1 + i\sqrt{2})^n + (1 - i\sqrt{2})^n\Big)$$

We could expand out the brackets using the binomial theorem but that would give us many terms with large binomial coefficients to have to work out. Instead we can use our understanding of complex multiplication to think about where in the plane these complex powers are. Calculating the polar representation we get that the modulus of each of $1 \pm i\sqrt{2}$ is $r = \sqrt{1+2} = \sqrt{3}$. The argument of $1 + i\sqrt{2}$ is $\theta = \arccos(1/\sqrt{3})$ while the argument of $1 - i\sqrt{2}$ is $-\arccos(1/\sqrt{3}) = -\theta$. De Moivre's theorem therefore immediately tells us that the n^{th} term is

$$\frac{1}{2}\left((\sqrt{3})^n(\cos(n\theta)+i\sin(n\theta))+(\sqrt{3})^n(\cos(n\theta)+i\sin(-n\theta))\right)=3^{n/2}\cos(n\theta).$$

6.2. **The cubic formula.** Recall the cubic formula for the solutions of the equation $x^3+cx+d=0$ are given by

$$x=\sqrt[3]{\frac{-d}{2}+\sqrt{\frac{d^2}{4}+\frac{c^3}{27}}}+\sqrt[3]{\frac{-d}{2}-\sqrt{\frac{d^2}{4}+\frac{c^3}{27}}}$$

We now have all the basic tools necessary to calculate the roots for any given c and d. The first thing to note is that cubic equations will generally have 3 solutions. This is reflected in the formula which involves cube roots, because after allowing for complex number solutions, numbers have three possible cube roots. At first sight it seems like the formula implies there could then be up to 9 solutions stemming from the choice of 3 cube roots for each of the two terms but if you follow the derivation, you see that the product of the two cube roots is equal to $-c/3$, i.e. a real number. In other words, if the argument of one is θ and the other is ϕ, we need $\sin(\theta+\phi)=0$, that is $\theta+\phi$ must be 0 or some other multiple of 2π.

De Moivre's theorem offers one way of calculating the cube roots explicitly. Suppose we want to find the three cube roots of an arbitrary complex number z written in polar form $z=r(\cos(\theta)+i\sin(\theta))$. It is straightforward to check using De Moivre's formula that each of the values

$$w=\begin{cases} r^{1/3}\left(\cos\left(\frac{\theta}{3}\right)+i\sin\left(\frac{\theta}{3}\right)\right)\\ r^{1/3}\left(\cos\left(\frac{\theta+2\pi}{3}\right)+i\sin\left(\frac{\theta+2\pi}{3}\right)\right)\\ r^{1/3}\left(\cos\left(\frac{\theta+4\pi}{3}\right)+i\sin\left(\frac{\theta+4\pi}{3}\right)\right)\end{cases}$$

satisfies $w^3=z$.

We can now work through Example 2 again without relying on knowing the answer ahead of time. Recall that the equation we wanted to solve was $x^3-15x-4=0$ and we used the cubic formula to write down the solutions as

$$x=\sqrt[3]{2+11i}+\sqrt[3]{2-11i}$$

Now to find the cube roots of $2+\sqrt{-121}$ as per the above suggestion we first calculate the modulus and argument

$$r=|2+11i|=\sqrt{2^2+11^2}=\sqrt{125}$$
$$\theta=\arg(2+11i)=\arctan(11/2)$$

and then use the formula above for cube roots. Doing likewise for $2-\sqrt{-121}$ and then applying the cubic formula (noting how the choice of first cube root implies the second) we get, after a little cancellation, the three roots

$$x=\begin{cases} 2\sqrt{5}\cos\left(\frac{\arctan(11/2)}{3}\right)=4\\ 2\sqrt{5}\cos\left(\frac{\arctan(11/2)+2\pi}{3}\right)=-3.7320508075688767\\ 2\sqrt{5}\cos\left(\frac{\arctan(11/2+4\pi)}{3}\right)=-0.26794919243112064\end{cases}$$

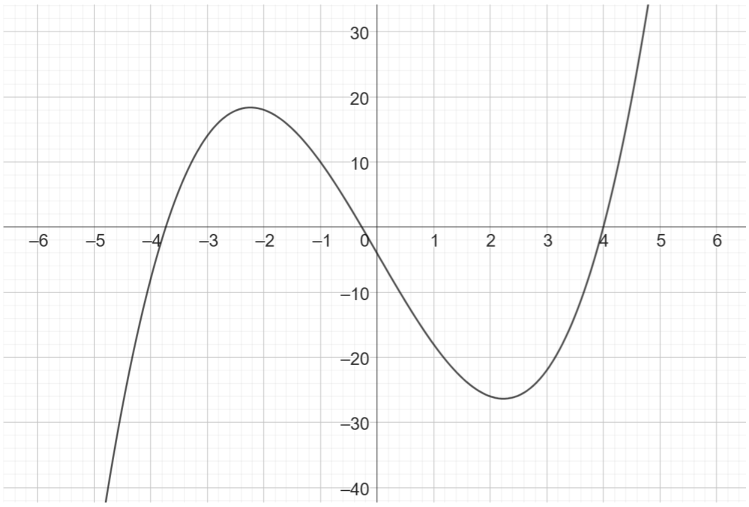


FIGURE 5. $y = x^3 - 15x - 4$

7. Euler's formula

In the previous section we promised a rather special formula. Let's pick up where we left off with De Moivre's Theorem in the form

$$\cos(\theta) + i\sin(\theta) = \left(\cos(\tfrac{\theta}{n}) + i\sin(\tfrac{\theta}{n})\right)^n$$

We saw that by comparing the real and imaginary parts of either side of this equation for different values of n, we get increasingly complicated trigonometric identities relating $\cos(\theta)$ and $\sin(\theta)$ to powers of $\cos(\frac{\theta}{n})$ and $\sin(\frac{\theta}{n})$. As n continues to increase without limit, the angle $\frac{\theta}{n}$ approaches 0. As a result, the approximations $\cos(\frac{\theta}{n}) \approx 1$ and $\sin(\frac{\theta}{n}) \approx \frac{\theta}{n}$ get more and more accurate and so too does the approximation

$$\cos(\theta) + i\sin(\theta) \approx \left(1 + i\tfrac{\theta}{n}\right)^n.$$

If we expand out the right hand side we get, using the binomial theorem,

$$1 + \frac{n}{n}\frac{(i\theta)}{1!} + \frac{n}{n}\frac{n-1}{n}\frac{(i\theta)^2}{2!} + \frac{n}{n}\frac{n-1}{n}\frac{n-2}{n}\frac{(i\theta)^3}{3!} + \cdots + \frac{n}{n}\frac{n-1}{n}\cdots\frac{n-k+1}{n}\frac{(i\theta)^k}{k!} + \cdots$$

For very large n, all the fractions of the form $\frac{n-j}{n}$ get closer to 1. (At least if j isn't too large with respect to n.) There is some subtlety around making these approximations precise but it isn't too hard[3] so in the end we're left with

$$\cos(\theta) + i\sin(\theta) = 1 + \frac{(i\theta)}{1!} + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \cdots + \frac{(i\theta)^k}{k!} + \cdots$$

As with finite n, we can compare real and imaginary parts of both sides using the fact that integer powers of i repeat with period 4

$$i^n = \begin{cases} 1 & \text{if } n = 4k \\ i & \text{if } n = 4k+1 \\ -1 & \text{if } n = 4k+2 \\ -i & \text{if } n = 4k+3 \end{cases}$$

[3] See Appendix D, for example.

$$\cos(\theta) = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots + (-1)^n \frac{\theta^{2n}}{(2n)!} + \cdots$$
$$\sin(\theta) = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} \cdots + (-1)^n \frac{\theta^{2n+1}}{(2n+1)!} + \cdots$$

Now the series for $\cos(\theta) + i\sin(\theta)$ is equal to the series for the exponential function $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!} + \cdots$ evaluated at $x = i\theta$ so we can also write it in the form

$$\cos(\theta) + i\sin(\theta) = e^{i\theta}$$

This relation is the famous 'Euler's formula'. It might seem strange to write $e^{i\theta}$. Maybe we could get used to some basic algebraic properties like addition and multiplication of numbers that square to negatives, but what could it possibly *mean* to raise a number to an imaginary power? As strange as it may seem, so are the identities we derived in Examples 3 and 4, and if we are to make any sense of those, we need a way of sensibly extending the meaning of functions like log, arctan, arcsin, etc. to complex input values.

It's quite analogous to how exponential notation is initially defined for the counting numbers and then extended to other kinds of numbers. The expression 2^6 is initially defined to mean 2 multiplied by itself 6 times. After introducing fractions and negative numbers, it is extremely convenient to extend the definition of exponents by assigning meaning to expressions such as $2^{10/3}$ and 2^{-7}. But now they do not in any straightforward sense mean 2 multiplied by itself $10/3$ or -7 times. Constrained by wanting to retain the identity $2^{x+y} = 2^x 2^y$, there is only one sensible way this extension can go. Assigning meaning to $2^{\sqrt{2}}$ is even less straightforward. There is no simple algebraic identity we can use, and again it's no good saying it means 2 multiplied by itself $\sqrt{2}$ times.

One approach is to note that for all rational values of x we have $2^x = e^{x\log(2)}$, and that we have a way of assigning meaning to the right hand side at least for all real values of x via the Taylor series. So for general real values of x like $\sqrt{2}$, we can define 2^x to mean $e^{x\log(2)}$ which in turn is defined by the infinite series. We can likewise extend the definition of any function with a Taylor series to arbitrary real and complex inputs (provided the series converges at the given input). This practice is so useful that it is very common to define functions via such series. As well as extending the definition to complex numbers, there are other advantages of this approach too. Using the fact that $\frac{d}{dx}\frac{x^n}{n!} = \frac{x^{n-1}}{(n-1)!}$ it easy to prove[4] using the infinite series that

$$\frac{d}{dx}e^x = e^x$$
$$\frac{d}{dx}\sin(x) = \cos(x)$$
$$\frac{d}{dx}\cos(x) = -\sin(x).$$

At least if we assume we are allowed to differentiate each term individually.

A disadvantage of using the series to define the exponential function e^x is that it's now not at all obvious that it satisfies the crucial identity $e^{x+y} = e^x e^y$ so that it legitimately extends the existing notation. Appendix E shows the connection between the series and this property.

[4] Exercise for the reader.

7.1. **Examples 3 and 4 revisited.** As discussed, we saw hints from Examples 3 and 4 that it might be interesting to consider functions like log with complex arguments. In particular, by treating i like any other number we derived the two identities

$$\arctan(x) = \frac{1}{2i}(\log(1+xi) - \log(1-xi))$$
$$\frac{1}{\sqrt{x^2-1}}\log(x+\sqrt{x^2-1}) = \frac{1}{\sqrt{1-x^2}}\arcsin(\sqrt{1-x^2})$$

We've seen how to assign meaning to $e^{i\theta}$ via the exponential series. There are a number of equivalent ways of defining the logarithm of a general complex number, all of which are a little fiddly for the following reason. We know $\log(x)$ is supposed to be the inverse of e^x but there is no standard or obvious way of assigning such an inverse because $e^{i\theta}$ is periodic with period 2π. For example, bearing in mind that $e^{i\pi/2} = e^{i5\pi/2} = i$, what should the logarithm of i be? $i\pi/2$, or $i5\pi/2$? The problem is very similar to (and is essentially a consequence of) not being able to assign an inverse to sin and cos that is valid for all inputs. One way of defining $\log(z)$ of complex number $z = r(\cos(\theta) + i\sin(\theta))$ given we know the value $\log(r)$ (since r is a positive real number) is

$$\log(z) = \log(r) + i\theta$$

where θ is chosen to be in the range $-\pi < \theta \leq \pi$. This indeed extends the definition of log as defined for positive real numbers because for such numbers, $\theta = 0$. It also continues to satisfy $e^{\log(x)} = x$ because, by Euler's formula,

$$e^{\log(z)} = e^{\log(r)+i\theta} = e^{\log(r)}e^{i\theta} = r(\cos(\theta) + i\sin(\theta)) = z.$$

Note that we used the property $e^{x+y} = e^x e^y$ which, again, is not obvious from the series definition but is not too hard to show[5].

Returning now to Example 3, recall that by evaluating the integral $\int_0^x \frac{1}{1+t^2}dt$ in two different ways we were lead to wonder about the possible equality

$$\arctan(x) = \frac{1}{2i}(\log(1+ix) - \log(1-ix)).$$

Multiplying both sides by $2i$ and taking the exponential of both sides, the identity becomes

$$e^{2i\arctan(x)} = \frac{1+ix}{1-ix}.$$

Using Euler's formula this equation is now not so mysterious after all. The left hand side is the complex number with modulus 1 and argument $2\arctan(x)$. But so is the right hand side. Here's one way to see that. The numerator $1+ix$ is the complex number with modulus $\sqrt{1+x^2}$ and argument $\arctan(x)$. The denominator has the same modulus $\sqrt{1+x^2}$ but argument $\arctan(-x)$. So when we divide them, the moduli cancel and the combined argument becomes $\arctan(x) - (-\arctan(x)) = 2\arctan(x)$, just as predicted by Euler's formula.

In Example 4 we considered the surface area of an ellipsoid. Recall that we had two different formulas depending on the shape of the underlying ellipse. We took the bold step of equation the two different formula and wrote down

$$\frac{1}{\sqrt{x^2-1}}\log(x+\sqrt{x^2-1}) = \frac{1}{\sqrt{1-x^2}}\arcsin(\sqrt{1-x^2})$$

[5] Appendix E.

Multiplying both sides by $\sqrt{1-x^2}\sqrt{x^2-1}$, replacing $\sqrt{x^2-1}$ with $i\sqrt{1-x^2}$ and taking exponentials of both sides the identity becomes

$$x + i\sqrt{1-x^2} = e^{i\arcsin(\sqrt{1-x^2})}$$

The link to Euler's identity should now be clear. If we let $\theta = \arcsin(\sqrt{1-x^2})$ then $x = \sqrt{1-\sin(\theta)^2} = \cos(\theta)$ and we are again left with the famous $e^{i\theta} = \cos(\theta) + i\sin(\theta)$.

8. Conclusion

We believe that the preceding calculations and explanations amount to good reasons for talking and thinking about $\sqrt{-1}$ as a new kind number that is real. To the readers that haven't felt its *ontological oomph*, we invite them not only to find alternatives, but to explain the success of the preceding ones.

References

[1] Paul J. Nahin, *An Imaginary Tale: The Story of* $\sqrt{-1}$

Appendix A. Recurrence formulas

In our first example we saw formulas for the n^{th} terms in the following pair of sequences

- $f_0 = 1,\ f_1 = 1$ and $f_{n+2} = f_{n+1} + f_n$

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

$$f_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

- $u_0 = 1,\ u_1 = 1$ and $u_{n+2} = 2u_{n+1} - 3u_n$

$$0, 1, 1, 2, 3, 5, 8, 13, 21, \ldots$$

$$u_n = \frac{1}{2}\left((1+\sqrt{-2})^n + (1-\sqrt{-2})^n\right).$$

Let's go through in a bit more detail why this these formula are true. The idea is the same for both except the second involves complex numbers where the first doesn't.

The first thing to check is that if x_1 and x_2 are the two roots of the quadratic $x^2 = x + 1$ then $f_n = x_1^n$ and $f_n = x_2$ both satisfy the relation $f_{n+2} = f_{n+1} + f_n$. You can also check that if a and b are any constants then $f_n = ax_1^n + bx_2^n$ also satisfies the same relation. The only remaining task, then, is to find values for a and b such that $f_0 = 0$ and $f_1 = 1$. So we need to solve the simultaneous equations in a and b

$$a + b = 0$$

$$ax_1 + bx_2 = 1$$

These give $a =$ and $b =$. The formula $f_n = ax_1^n + bx_2^n$ with these values of a and b therefore satisfies all the defining equations for f_n and is therefore equal to the given sequence.

Exactly the same procedure results in the formula above for u_n. We leave the details as an exercise for the interested reader to complete.

Appendix B. Cardano's cubic formula

We're facing the equation

$$x^3 + cx + d = 0$$

and we're trying to find a formula for x in terms of c and d. Start by writing $x = y + z$ so that we have two unknowns y and z and the equation becomes

$$y^3 + z^3 + d + (y + z)(3yz + c) = 0.$$

It is not at all obvious why we would do this but now that we have, the idea for how we might go about solving the equation is easy enough to explain. We will try to find y and z that simultaneously solve the two equations $y^3 + z^3 + d = 0$ and $3yz + c = 0$. Using the second to solve for z and substituting into the first we have $z = -c/(3y)$ and $y^3 + (-c/(3y))^3 + d = 0$ or, after multiplying by y^3,

$$y^6 + dy^3 - c^3/27 = 0.$$

This might look even harder than the original equation because it is degree 6 but notice that it is really a quadratic in disguise in y^3 which we can solve using the quadratic formula.

$$y^3 = \frac{-d \pm \sqrt{d^2 + 4c^3/27}}{2}.$$

Now if $y^3 = \frac{-d+\sqrt{d^2+4c^3/27}}{2}$ then $z^3 = \frac{-d-\sqrt{d^2+4c^3/27}}{2}$ and vice versa so altogether, after taking the cube root, we have found a solution to our original equation in x at it is given by

$$x = \sqrt[3]{\frac{-d}{2} + \sqrt{\frac{d^2}{4} + \frac{c^3}{27}}} + \sqrt[3]{\frac{-d}{2} - \sqrt{\frac{d^2}{4} + \frac{c^3}{27}}}.$$

Appendix C. Two integrals

In this appendix we shall derive the following results

$$I(x) = \int_0^x \sqrt{1-u^2}du = \frac{x\sqrt{1-x^2}}{2} - \frac{\arcsin(x)}{2}$$

$$J(x) = \int_0^x \sqrt{1+u^2}du = \frac{x\sqrt{1+x^2}}{2} + \frac{\log(x+\sqrt{1+x^2})}{2}$$

First the first, we can use the substitution $u = \sin(t)$. Then $du/dt = \cos(t)$ and $\sqrt{1-\sin(t)^2} = \cos(t)$ so that

$$I(x) = \int_0^{\arcsin(x)} \cos(t)^2 dt.$$

Using the identity $\cos(t)^2 = \frac{1}{2}\cos(2t) - \frac{1}{2}$

$$I(x) = \int_0^{\arcsin(x)} \left(\frac{1}{2}\cos(2t) - \frac{1}{2}\right) dt = \left[\frac{1}{4}\sin(2t) - \frac{t}{2}\right]_0^{\arcsin(x)}.$$

Then using the double angle formula $\sin(2t) = 2\sin(t)\cos(t)$ together with $\cos(t) = \sqrt{1-\sin(t)^2}$ again this is equal to

$$I(x) = \frac{x\sqrt{1-x^2}}{2} - \frac{\arcsin(x)}{2}$$

as required.

Now for the second, we can start by using the substitution $u = \tan(t)$. Then $du/dt = 1/\cos(t)^2$ and $\sqrt{1+\tan(t)^2} = 1/\cos(t)$ so that

$$J(x) = \int_0^{\arctan(x)} \frac{1}{\cos(t)^3} dt.$$

Integrating $1/\cos(t)^2$ by parts to get $\tan(t)$ and differentiating the other factor $1/\cos(t)$ to get $\frac{\sin(t)}{\cos(t)^2}$ this is

$$\begin{aligned}
J(x) &= \left[\tan(t)\frac{1}{\cos(t)}\right]_{t=0}^{t=\arctan(x)} - \int_0^{\arctan(x)} \frac{\tan(t)\sin(t)}{\cos(t)^2} dt \\
&= x\sqrt{1+x^2} + \int_0^{\arctan(x)} \frac{\sin(t)^2}{\cos(t)^3} dt \\
&= x\sqrt{1+x^2} - \int_0^{\arctan(x)} \frac{1-\cos(t)^2}{\cos(t)^3} dt \\
&= x\sqrt{1+x^2} - J(x) + \int_0^{\arctan(x)} \frac{1}{\cos(t)} dt.
\end{aligned}$$

Now $J(x)$ appears on both sides and rearranging we have

$$J(x) = \frac{x\sqrt{1+x^2}}{2} + \int_0^{\arctan(x)} \frac{1}{\cos(t)} dt.$$

This last integral is also not entirely straightforward but can be done by writing it in a form $f'(t)/f(t)$ where $f(t) = 1/\cos(t) + \sin(t)/\cos(t)$ which we can integrate to $\log(f(t))$

$$\int_0^{\arctan(x)} \frac{1/\cos(t)^2 + \sin(t)/\cos(t)^2}{1/\cos(t) + \sin(t)/\cos(t)} dt = [\log(1/\cos(t) + \sin(t)/\cos(t))]_{t=0}^{t=\arctan(x)} = \log(x+\sqrt{1+x^2}).$$

Appendix D. Power series for $\cos + i\sin$

We want to show that $\cos(\theta) + i\sin(\theta) = \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!}$. It suffices to show both of the following

(1) Show that $\lim_{n\to\infty} \left(1 + i\frac{\theta}{n}\right)^n = \cos(\theta) + i\sin(\theta)$
(2) Show that $\lim_{n\to\infty} \left(1 + i\frac{\theta}{n}\right)^n = \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!}$

To prove (1) we can calculate the modulus and argument of $\left(1 + i\frac{\theta}{n}\right)^n$ and show that they approach 1 and θ, respectively, as n goes to infinity. These two lemmas which can themselves be proved easily are enough for that. (2) can be proved by multiplying out the brackets and using the second lemma again. REMAINING DETAILS TO BE ADDED.

Lemma. *For $\theta > 0$ we have* $\sin(\theta) \le \theta \le \tan(\theta)$.

Lemma. *For any positive integer n and any $0 < x \le n^2$ we have* $1 - x/n \le (1 - x/n^2)^n$.

Proof. Let $y = 1 - x/n^2$. Then $n \ge 1 + y + y^2 + \cdots y^{n-1} = \frac{1-y^n}{1-y} = \frac{1-(1-x/n^2)^n}{x/n^2}$. □

Appendix E. Addition property for exp

In this appendix we will see how the property

$$\exp(x+y) = \exp(x)\exp(y)$$

follows from the series definition

$$\exp(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots + \frac{x^n}{n!} + \cdots$$

We can work directly with the series using the binomial theorem

$$\begin{aligned} \exp(x+y) &= \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!} \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n!} x^k y^{n-k} = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{x^k}{k!} \frac{y^{n-k}}{(n-k)!} = \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} \frac{x^k}{k!} \frac{y^{n-k}}{(n-k)!} \\ &= \sum_{k=0}^{\infty} \frac{x^k}{k!} \sum_{m=0}^{\infty} \frac{y^m}{m!} = \exp(x)\exp(y). \end{aligned}$$